Artificial intelligence

Machine learning and its types

Learning

In AI, "learning" refers to how AI systems improve their ability to perform tasks or make predictions over time. This is achieved through various techniques, including

- Supervised learning
- Unsupervised learning
- Reinforcement learning

Supervised learning

Supervised learning is defined as when a model gets trained on a **"Labelled Dataset"**. Labelled datasets have both input and output parameters. In **Supervised Learning** algorithms learn to map points between inputs and correct outputs. It has both training and validation datasets labelled.

There are two main categories of supervised learning that are mentioned below:

- Classification
- Regression

Classification

Deals with predicting **categorical** target variables, which represent discrete classes or labels. For instance, classifying emails as spam or not spam, or predicting whether a patient has a high risk of heart disease. Classification algorithms learn to map the input features to one of the predefined classes.

Here are some classification algorithms:

- Logistic Regression
- Support Vector Machine
- Random Forest
- Decision Tree
- K-Nearest Neighbors (KNN)
- Naive Bayes

Regression

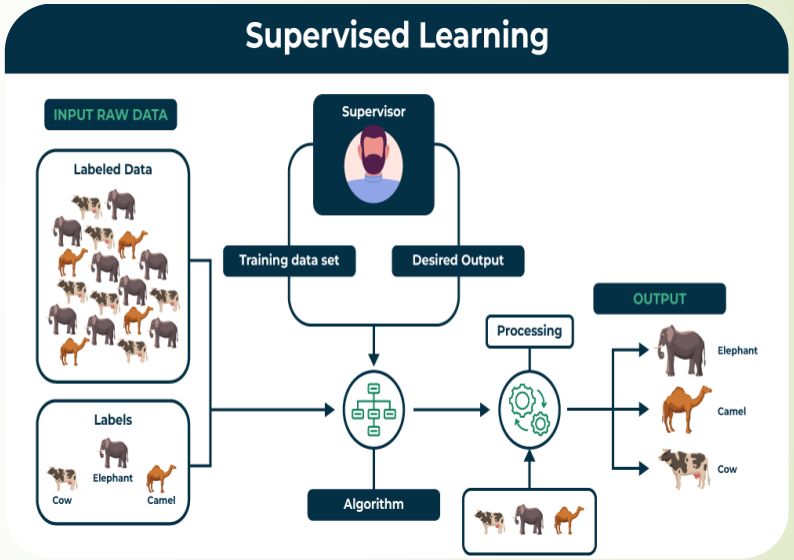
Deals with predicting **continuous** target variables, which represent numerical values. For example, predicting the price of a house based on its size, location, and amenities, or forecasting the sales of a product. Regression algorithms learn to map the input features to a continuous numerical value.

Here are some regression algorithms:

- Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression
- Decision tree
- Random Forest

Example of supervised learning

Consider a scenario where you have to build an image classifier to differentiate between cats and dogs. If you feed the datasets of dogs and cats labelled images to the algorithm, the machine will learn to classify between a dog or a cat from these labeled images. When we input new dog or cat images that it has never seen before, it will use the learned algorithms and predict whether it is a dog or a cat.



Applications

- **Image classification**: Identify objects, faces, and other features in images.
- **Natural language processing:** Extract information from text, such as sentiment, entities, and relationships.
- **Speech recognition**: Convert spoken language into text.
- **Recommendation systems**: Make personalized recommendations to users.
- **Predictive analytics**: Predict outcomes, such as sales, customer churn, and stock prices.
- **Medical diagnosis**: Detect diseases and other medical condition

Pros and cons

Pros:

- Supervised Learning models can have high accuracy as they are trained on labelled data.
- The process of decision-making in supervised learning models is often interpretable.
- It can often be used in pre-trained models which saves time and resources

Cons:

- It has limitations in knowing patterns.
- It can be time-consuming and costly as it relies on **labeled** data only.
- It may lead to poor generalizations based on new data.

Unsupervised Machine Learning

Unsupervised learning is a type of machine learning technique in which an algorithm discovers patterns and relationships using unlabeled data

The primary goal of Unsupervised learning is often to discover hidden patterns, similarities, or clusters within the data, which can then be used for various purposes, such as data exploration, visualization, dimensionality reduction, and more.

There are two main categories of unsupervised learning that are mentioned below:

- Clustering
- Association

Clustering

The process of grouping data points into clusters based on their similarity. This technique is useful for identifying patterns and relationships in data without the need for labeled examples.

Here are some clustering algorithms:

- K-Means Clustering algorithm
- Mean-shift algorithm
- DBSCAN Algorithm
- Principal Component Analysis
- Independent Component Analysis

Association

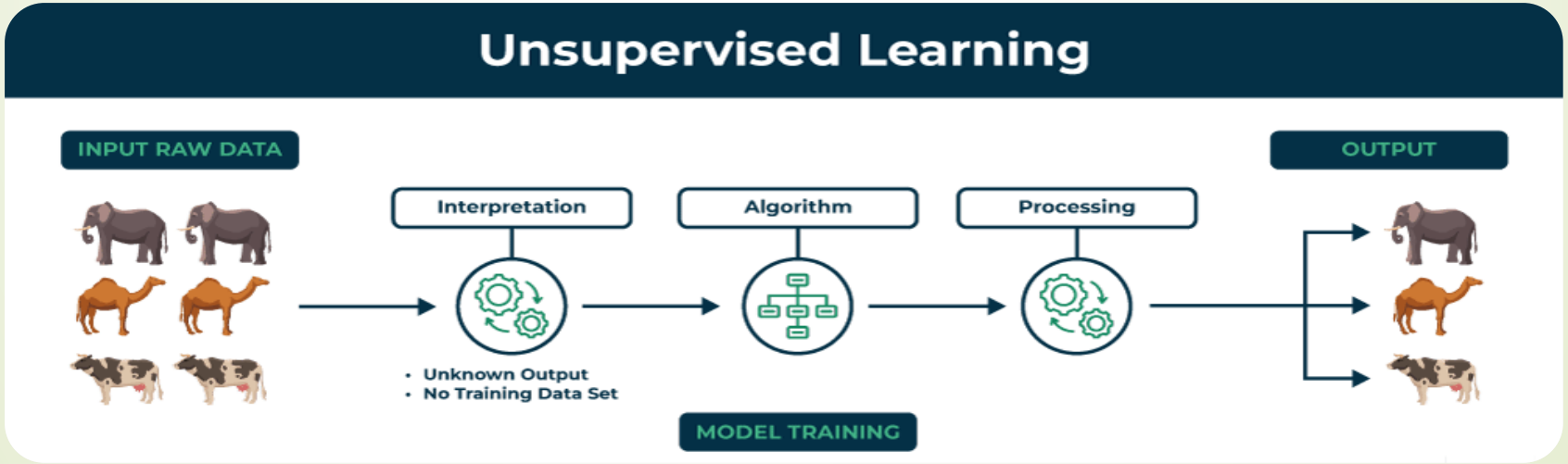
A technique for discovering relationships between items in a dataset. It identifies rules that indicate the presence of one item implies the presence of another item with a specific probability.

Here are some association rule learning algorithms:

- Apriori Algorithm
- Eclat
- FP-growth Algorithm

Example

Consider that you have a dataset that contains information about the purchases you made from the shop. Through clustering, the algorithm can group the same purchasing behavior among you and other customers, which reveals potential customers without predefined labels. This type of information can help businesses get target customers as well as identify out.



Applications

- **Clustering**: Group similar data points into clusters.
- **Anomaly detection**: Identify outliers or anomalies in data.
- **Dimensionality reduction**: Reduce the dimensionality of data while preserving its essential information.
- **Recommendation systems**: Suggest products, movies, or content to users based on their historical behavior or preferences.
- **Topic modeling**: Discover latent topics within a collection of documents.
- **Density estimation**: Estimate the probability density function of data.

Pros and cons

Pros:

- It helps to discover hidden patterns and various relationships between data.
- Used for tasks such as customer segmentation, anomaly detection, and data exploration.
- It does not require labeled data and reduces the effort of data labeling.

Cons:

- Without using labels, it may be difficult to predict the quality of model's output.
- It has techniques such as_auto encoder sand dimensionality reduction that can be used to extract meaningful features from raw data.

Reinforcement learning

Reinforcement machine learning algorithm is a learning method that interacts with the environment by producing actions and discovering errors.

Trial, error, and delay are the most relevant characteristics of reinforcement learning.

In this technique, the model keeps on increasing its performance using Reward Feedback to learn the behavior or pattern.

These algorithms are specific to a particular problem e.g. Google Self Driving car, Alpha Go where a bot competes with humans and even itself to get better and better performers in Go Game.

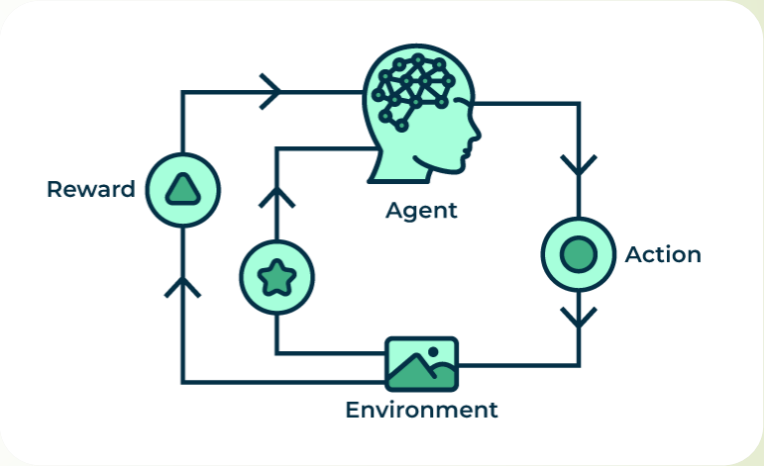
Reinforcement algorithms

Here are some of most common reinforcement learning algorithms:

- **Q-learning :** Q-learning is a model-free RL algorithm that learns a Q-function, which maps states to actions. The Q-function estimates the expected reward of taking a particular action in a given state.
- **SARSA (State-Action-Reward-State-Action):** SARSA is another model-free RL algorithm that learns a Q-function. However, unlike Q-learning, SARSA updates the Q-function for the action that was actually taken, rather than the optimal action.
- **Deep Q-learning:** Deep Q-learning is a combination of Q-learning and deep learning. Deep Q-learning uses a neural network to represent the Q-function, which allows it to learn complex relationships between states and actions

Example

Consider that you are training an AI agent to play a game like chess. The agent explores different moves and receives positive or negative feedback based on the outcome. Reinforcement Learning also finds applications in which they learn to perform tasks by interacting with their surroundings.



Types of reinforcement

Positive reinforcement:

- Rewards the agent for taking a desired action.
- Encourages the agent to repeat the behavior.
- Examples: Giving a treat to a dog for sitting, providing a point in a game for a correct answer.

Negative reinforcement:

- Removes an undesirable stimulus to encourage a desired behavior.
- Discourages the agent from repeating the behavior.
- Examples: Turning off a loud buzzer when a lever is pressed, avoiding a penalty by completing a task.

Applications

- **Game Playing**: RL can teach agents to play games, even complex ones.
- **Robotics**: RL can teach robots to perform tasks autonomously.
- **Autonomous Vehicles**: RL can help self-driving cars navigate and make decisions.
- **Recommendation Systems**: RL can enhance recommendation algorithms by learning user preferences.
- **Healthcare**: RL can be used to optimize treatment plans and drug discovery.
- **Natural Language Processing (NLP)**: RL can be used in dialogue systems and chat bots.

Pros and cons

Pros:

- It has autonomous decision-making that is well-suited for tasks and that can learn to make a sequence of decisions, like robotics and game-playing.
- This technique is preferred to achieve long-term results that are very difficult to achieve.
- It is used to solve a complex problems that cannot be solved by conventional techniques.

Cons:

- Training Reinforcement Learning agents can be computationally expensive and time-consuming.
- Reinforcement learning is not preferable to solving simple problems.
- It needs a lot of data and a lot of computation, which makes it impractical and costly.